PREOPERATIVE DIAGNOSIS: Complete rupture of the right anterior cruciate ligament with associated complex tear of the posterior horn of the medial meniscus.

POSTOPERATIVE DIAGNOSIS: Complete ACL rupture; complex tear medial meniscus.

PROCEDURE: Right knee arthroscopy, partial medial meniscectomy, and anatomic single-bundle bone-patellar tendon-bone (BTB) autograft ACL reconstruction.

PROCEDURE SUMMARY: The patient was placed in the supine position under general anesthesia with a left thigh lateral post and well-padded pneumatic tourniquet inflated to 260 mmHg. Anterolateral and anteromedial portals were established. Diagnostic arthroscopy confirmed a complete midsubstance tear of the anterior cruciate ligament with typical anterior translation on pivot-shift testing. Evaluation of the medial compartment revealed a complex degenerative maceration involving the posterior horn of the medial meniscus. Using a combination of straight and curved basket punch forceps and motorized shaver, a stable partial medial meniscectomy was executed back to a vascular bleeding rim.

Graft Harvest: A longitudinal midline incision from the inferior patellar pole to the tibial tubercle was created. A central 10 mm wide bone-patellar tendon-bone autograft was harvested with matching 20 mm trapezoidal patellar and tibial bone plugs using an oscillating saw and osteotomes. The bone plugs were contoured and drilled with passing sutures. The patellar defect was grafted with cancellous bone chips, and the peritenon was closed with 2-0 Vicryl.

Femoral and Tibial Tunnel Preparation: Through an accessory anteromedial portal with the knee flexed to 120 degrees, an anatomic femoral tunnel was drilled to a depth of 25 mm using a 10 mm cannulated drill bit positioned at the native femoral footprint (1:30 clock-face position). A tibial guide set at 55 degrees was placed through the anteromedial portal onto the tibial ACL footprint, and a 10 mm cannulated tibial tunnel was drilled over a guide pin.

Graft Passage and Fixation: The BTB autograft was passed through the tibial tunnel into the femoral tunnel under direct arthroscopic visualization. Femoral fixation was achieved utilizing a 7 x 20 mm titanium interference screw flush with the cortex. The knee was cycled through 30 full flexion-extension cycles under maximal manual graft tension. Tibial fixation was secured with an 8 x 25 mm bioabsorbable interference screw at 20 degrees of knee flexion with a posterior drawer maneuver applied. Arthroscopic inspection verified full knee extension without notch impingement and negative anterior drawer and Lachlan tests. Standard layered closure was executed.